

Notation / Style

Largely following the Boyd Convex Optimisation textbook

The natural numbers $\mathbb{N}$ are $\{0, 1, 2, 3, \dots\}$ and $\mathbb{N}^* = \mathbb{N} \setminus \{0\}$.

Let $p, q \in \mathbb{N}$.

The typical ambient space is $\mathbb{R}^p$. Functions are generally $\mathbb{R}^p \rightarrow \mathbb{R}^q$. The function definition is specified in the ‘computer’ style:

$f : \mathbb{R}^p \rightarrow \mathbb{R}^q$ describes the input/output spaces
and $\text{dom } f$ or $\mathcal{D}$ denote the domain of definition of $f(x)$

If $x \in \mathbb{R}^p$, then without further information $x = (x_k)$, i.e., $x_k, k = 1, \dots, p$ are usually its elements. k is the first choice but, if other indices are required then j, n, m, i are used (descending order of preference). Prefer n, k if limits are taken.

A typical set in $\mathbb{R}^p$ is A, B (rather than C). Prefer $\subseteq$ rather than $\subset$ unless the strict inclusion is important.